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REAGENT AND MECHANISM

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Reagents

1. What is the decreasing order of strength of the bases $OH^{\ominus}$, $NH_2^{\ominus}$, $HC\equiv C^{\ominus}$ and $CH_3-CH_2^{\ominus}$?

(a) $CH_3CH_2^{\ominus} > NH_2^{\ominus} > HC\equiv C^{\ominus} > OH^{\ominus}$

(b) $CH\equiv C^{\ominus} > CH_3CH_2^{\ominus} > NH_2^{\ominus} > OH^{\ominus}$

(c) $OH^{\ominus} > NH_2^{\ominus} > HC\equiv C^{\ominus} > CH_3CH_2^{\ominus}$

(d) $NH_2^{\ominus} > HC\equiv C^{\ominus} > OH^{\ominus} > CH_3CH_2^{\ominus}$

2. $CH_3^{\ominus}$, $NH_2^{\ominus}$, $OH^{\ominus}$ and $F^{\ominus}$ in increasing pK_b values are :

(a) $CH_3^{\ominus} < NH_2^{\ominus} < OH^{\ominus} < F^{\ominus}$

(b) $F^{\ominus} < OH^{\ominus} < NH_2^{\ominus} < CH_3^{\ominus}$

(c) $OH^{\ominus} < NH_2^{\ominus} < CH_3^{\ominus} < F^{\ominus}$

(d) None of correct

3. Which of the following species is an electrophile ?

(a) $H^{\oplus}$ (b) $H^{\ominus}$

(c) both (a) and (b) (d) none of these

4. Which of the following species is a nucleophile ?

(a) H_2O (b) $H_3O^{\oplus}$

(c) both (a) and (b) (d) none of these

5. Which of the following species is a nucleophile ?

(a) $CH_3O^{\ominus}$ (b) CH_3OH

(c) both (a) and (b) (d) none of these

6. $\overset{\ominus}{C}H_3$, $\overset{\ominus}{N}H_2$, $\overset{\ominus}{O}H$, $\overset{\ominus}{F}$

Among these nucleophiles, which of the following orders is correct for their nucleophilicity order in gas phase ?

(a) $\overset{\ominus}{C}H_3 > \overset{\ominus}{N}H_2 > \overset{\ominus}{O}H > \overset{\ominus}{F}$ (b) $\overset{\ominus}{F} > \overset{\ominus}{O}H > \overset{\ominus}{N}H_2 > \overset{\ominus}{C}H_3$

(c) $\overset{\ominus}{O}H > \overset{\ominus}{N}H_2 > \overset{\ominus}{F} > \overset{\ominus}{C}H_3$ (d) none of these

7. $\overset{\ominus}{C}H_3$, $\overset{\ominus}{N}H_2$, $\overset{\ominus}{O}H$, $\overset{\ominus}{F}$

Among these nucleophiles, which of the following orders is correct for their nucleophilicity in acetone ?

(a) $\overset{\ominus}{C}H_3 > \overset{\ominus}{N}H_2 > \overset{\ominus}{O}H > \overset{\ominus}{F}$ (b) $\overset{\ominus}{F} > \overset{\ominus}{O}H > \overset{\ominus}{N}H_2 > \overset{\ominus}{C}H_3$

(c) $\overset{\ominus}{O}H > \overset{\ominus}{N}H_2 > \overset{\ominus}{F} > \overset{\ominus}{C}H_3$ (d) none of these

8. $F^{\ominus}, Cl^{\ominus}, Br^{\ominus}, I^{\ominus}$

Among these nucleophiles, which of the following orders is correct for their nucleophilicity order in DMF ?

(a) $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$ (b) $I^{\ominus} > Br^{\ominus} > Cl^{\ominus} > F^{\ominus}$

(c) $I^{\ominus} > Br^{\ominus} > F^{\ominus} > Cl^{\ominus}$ (d) none of these

9. $F^{\ominus}, Cl^{\ominus}, Br^{\ominus}, I^{\ominus}$

Among these nucleophiles, which of the following orders is correct for their nucleophilicity order in water ?

(a) $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$ (b) $I^{\ominus} > Br^{\ominus} > Cl^{\ominus} > F^{\ominus}$

(c) $I^{\ominus} > Br^{\ominus} > F^{\ominus} > Cl^{\ominus}$ (d) none of these

10. For the following :

(1) I^- (2) Cl^- (3) Br^-

the increasing order of nucleophilicity in polar protic solvent would be

(a) $I^- < Br^- < Cl^-$ (b) $Cl^- < Br^- < I^-$

(c) $I^- < Cl^- < Br^-$ (d) $Br^- < Cl^- < I^-$

Substitution

11. The less reactivity of chlorine atom in $CH_2=CH-Cl$ is due to

(a) Inductive effect (b) Resonance stabilization

(c) Electromeric effect (d) Electronegativity

12. $R-X + NaOH \longrightarrow ROH + NaX$

The above reaction is classified as

(a) Nucleophilic substitution

(b) Electrophilic substitution

(c) Reduction (d) Oxidation

13. Among the following, the one which reacts most readily with ethanol is

(a) *p*-nitrobenzyl bromide

(b) *p*-chlorobenzyl bromide

(c) *p*-methoxybenzyl bromide

(d) *p*-methylbenzyl bromide

14. Isopropyl chloride undergoes hydrolysis by :

(a) S_N1 mechanism (b) S_N2 mechanism

(c) S_N1 and S_N2 mechanism

(d) Neither S_N1 nor S_N2 mechanism

15. Which one of the following is most reactive towards nucleophilic substitution ?

(a) $H_2C{=}CH{-}Cl$ (b) C_6H_5Cl

(c) $CH_3{-}CH{=}CH{-}Cl$

(d) $ClCH_2{-}CH{=}CH_2$

16. Which of the following is the example of S_N2 reaction

(a) $CH_3Br + OH^{\ominus} \longrightarrow CH_3OH + Br^{\ominus}$

(b) $CH_3{-}\underset{\underset{Br}{|}}{CH}{-}CH_3 + OH^{\ominus} \longrightarrow CH_3\underset{\underset{OH}{|}}{CH}CH_3 + Br^{\ominus}$

(c) $CH_3CH_2OH \xrightarrow{-H_2O} CH_2{=}CH_2$

(d) $CH_3{-}\overset{\overset{CH_3}{|}}{\underset{\underset{Br}{|}}{C}}{-}CH_3 + OH^{\ominus} \longrightarrow CH_3{-}\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}{-}OH + Br^{\ominus}$

17. In S_N1 reaction, the racemization takes place. It is due to

(a) conversion of configuration

(b) retention of configuration

(c) inversion of configuration

(d) both (b) & (c)

18. Arrange the following halides in the decreasing order of S_N1 reactivity

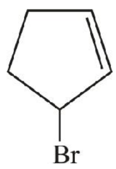
(I) $CH_3CH_2CH_2Cl$ (II) $CH_2{=}CHCH(Cl)CH_3$

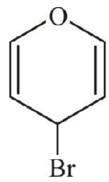
(III) $CH_3CH_2CH(Cl)CH_3$

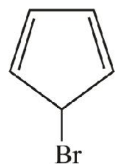
(a) I > II > III (b) II > III > I

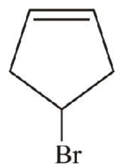
(c) II > I > III (d) III > II > I

19. The increasing order of reactivity of the following bromides in S_N1 reaction is

(I) 

(II) 

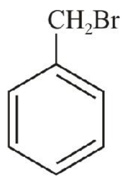
(III) 

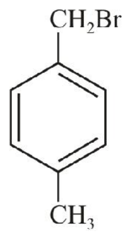
(IV) 

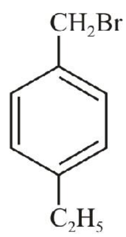
(a) III > I > II > IV (b) III > II > I > IV

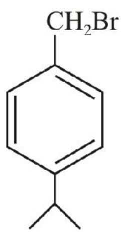
(c) II > III > I > IV (d) II > I > IV > III

20. Arrange the following halides in decreasing order of reactivity in S_N1 reaction

(I) 

(II) 

(III) 

(IV) 

(a) II > III > IV > I (b) IV > III > II > I

(c) III > IV > II > I (d) I > II > III > IV

21. Which of the following two reactions would be faster?

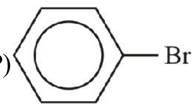
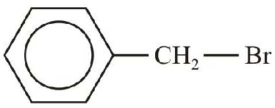
(a) $CH_3-I+\overset{\ominus}{O}H \rightarrow CH_3-OH+\overset{\ominus}{I}$

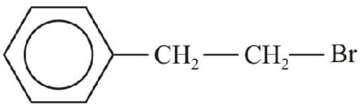
(b) $CH_3-Cl+\overset{\ominus}{O}H \rightarrow CH_3-OH+\overset{\ominus}{Cl}$

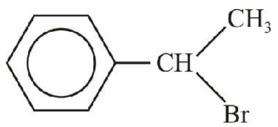
(c) both have similar rates

(d) rates cannot be predicted

22. Rate of S_N1 reaction is:

(P)  (Q) 

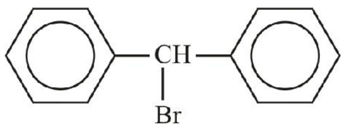
(R) 

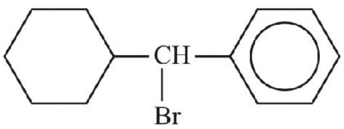
(S) 

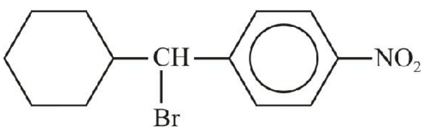
(a) S > Q > R > P (b) S > R > P > Q

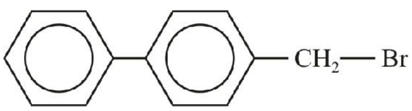
(c) P > Q > R > S (d) S > R > Q > P

23. The rate of S_N1 reaction is fastest with:

(a) 

(b) 

(c) 

(d) 

24. $H_3C-\overset{\ominus}{S}\,\overset{\oplus}{Na}+CH_3CH_2-X \rightarrow$

The reaction is fastest when X is:

(a) – OH (b) – F

(c) $-O-\overset{O}{\overset{\|}{C}}-CF_3$ (d) $-O-\overset{O}{\overset{\|}{C}}-CH_3$

25. When the concentration of alkyl halide is tripled and concentration of $\overset{\ominus}{O}H$ is reduced to half, the rate of S_N2 reaction increased by:

(a) 3 times (b) 1.5 times

(c) 2 times (d) 6 times

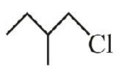
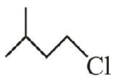
26. Arrange the following in decreasing order of S_N2 reaction (From question 26 to 30)

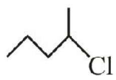
(P) CH_3Cl (Q) CH_3CH_2Cl

(R) $CH_3CH_2CH_2Cl$ (S) $CH_3-\underset{}{\overset{Cl}{\overset{|}{C}H}}-CH_3$

(a) P > Q > S > R (b) P > Q > R > S

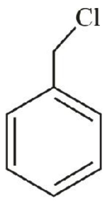
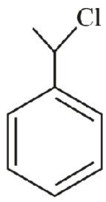
(c) S > R > Q > P (d) S > > R > P

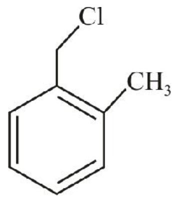
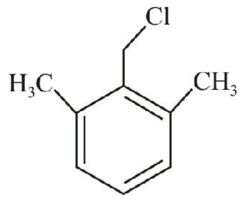
27. (P)  (Q) 

(R)  (S) 

(a) P > Q > R > S (b) R > P > Q > S

(c) Q > R > P > S (d) R > Q > P > S

28. (P)  (Q) 

(R)  (S) 

(a) P > R > S > Q (b) P > Q > R > S

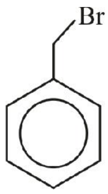
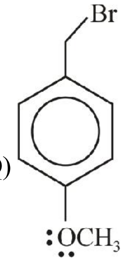
(c) P > R > Q > S (d) R > Q > S > P

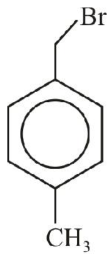
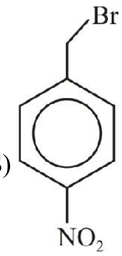
29. (P)  (Q) 

(R) H_3C-Br (S) $H_2C=CH-Br$

(a) S > P > Q > R (b) Q > S > R > P

(c) Q > R > P > S (d) R > Q > P > S

30. (P)  (Q) 

(R)  (S) 

(a) Q > R > P > S (b) R > Q > S > P

(c) P > Q > R > S (d) S > P > R > Q

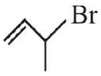
31. Arrange the following in decreasing order of reactivity towards S_N1 reaction (Q 31 to 35)

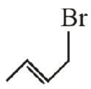
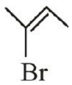
(P) $CH_3—Cl$ (Q) 

(R)  (S) 

(a) S > R > Q > P (b) S > Q > R > P

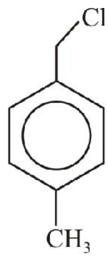
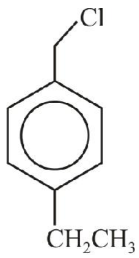
(c) R > S > Q > P (d) P > Q > R > S

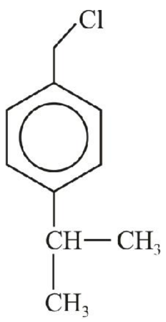
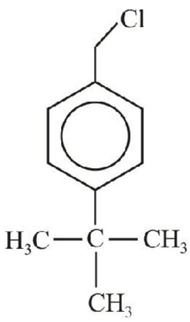
32. (P)  (Q) 

(R)  (S) 

(a) P > Q > R > S (b) Q > R > S > P

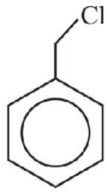
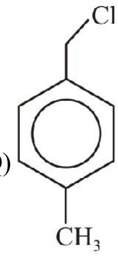
(c) Q > P > R > S (d) Q > R > P > S

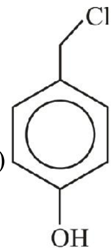
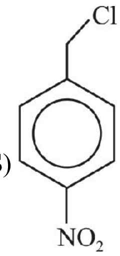
33. (P)  (Q) 

(R)  (S) 

(a) S > R > Q > P (b) R > S > Q > P

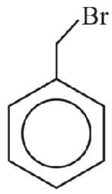
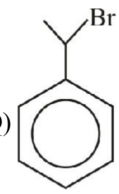
(c) Q > R > S > P (d) P > Q > R > S

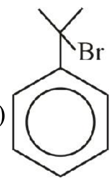
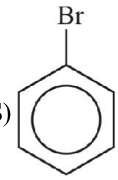
34. (P)  (Q) 

(R)  (S) 

(a) S > P > Q > R (b) R > Q > P > S

(c) Q > R > P > S (d) S > Q > P > R

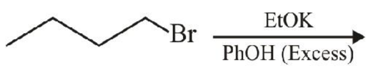
35. (P)  (Q) 

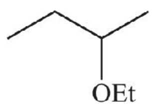
(R)  (S) 

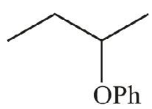
(a) P > Q > R > S (b) S > P > Q > R

(c) R > Q > P > S (d) R > P > Q > S

36. Predict the major product of the reaction



(a)  (b) 

(c)  (d) 

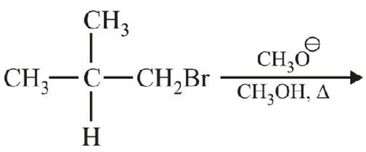
Elimination

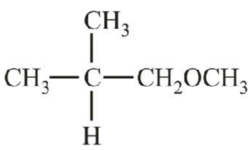
37. 2-bromopentane is heated with potassium ethoxide in ethanol. The major product obtained is

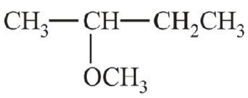
(a) Pent-1-ene (b) *cis* pent-2-ene

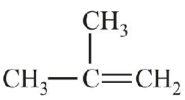
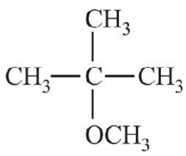
(c) *trans* pent-2-ene (d) 2-ethoxypentane

38. The major product formed in the following reaction is



(a) 

(b) 

(c)  (d) 

39. The major product obtained on treatment of $CH_3CH_2CH(F)CH_3$ with CH_3O^-/ CH_3OH is

(a) $CH_3CH_2CH(OCH_3)\ CH_3$

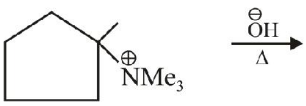
(b) $CH_3CH{=}CHCH_3$

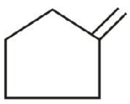
(c) $CH_3CH_2CH{=}CH_2$ (d) $CH_3CH_2CH_2CH_2OCH_3$

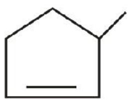
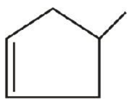
40. n-Propyl bromide on treatment with ethanolic potassium hydroxide produces

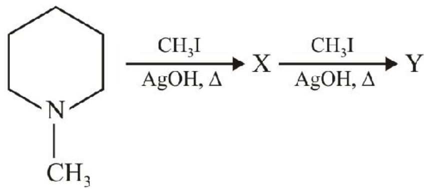
(a) Propane (b) Propene

(c) Propyne (d) Propanol

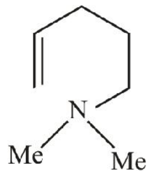
41.  Major product :

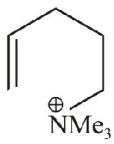
(a)  (b) 

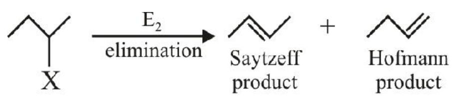
(c)  (d) 

42. 

Identify the major product Y:

(a)  (b) 

(c)  (d) 

43. 

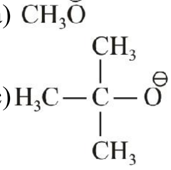
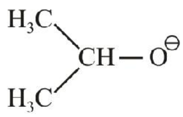
In the above reaction, maximum Saytzeff product will be obtained where X is:

(a) I (b) Cl

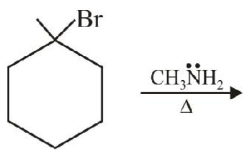
(c) Br (d) F

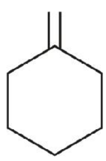
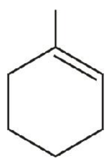
44. In the above reaction Hoffman product is major when base is:

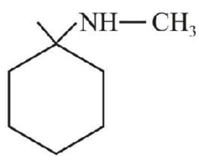
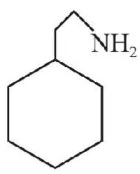
(a) $CH_3\overset{\ominus}{O}$ (b) $CH_3CH_2\overset{\ominus}{O}$

(c)  (d) 

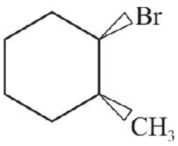
45. Major product of the following reaction is:

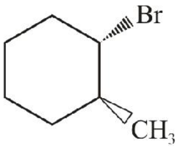


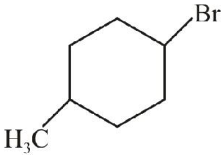
(a)  (b) 

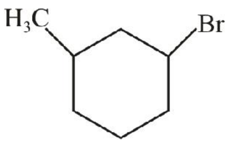
(c)  (d) 

46. Which of the following will undergo fastest elimination with alcoholic KOH?

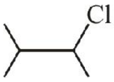
(a) 

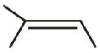
(b) 

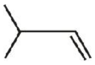
(c) 

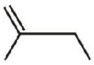
(d) 

47. Find the major product of the reaction:

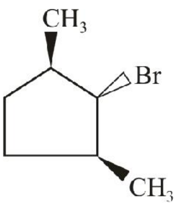
 $\xrightarrow{Me_3CO^{\ominus}K^{\oplus}}$

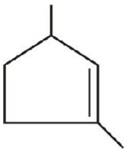
(a) 

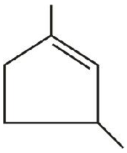
(b) 

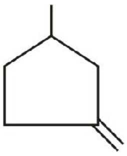
(c) 

(d) Both (a) and (b)

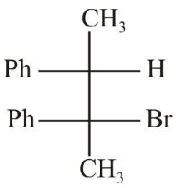
48.  $\xrightarrow[CH_3OH]{CH_3\overset{\ominus}{O}\overset{\oplus}{Na}}$ Product :

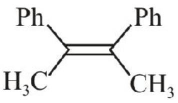
(a) 

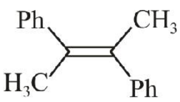
(b) 

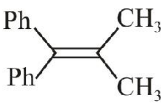
(c) 

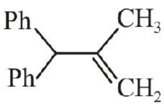
(d) No reaction

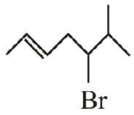
49.  $\xrightarrow{\text{alc. KOH}}$ Major product :

(a) 

(b) 

(c) 

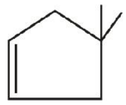
(d) 

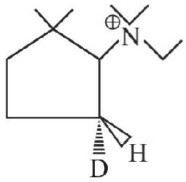
50.  $\xrightarrow{\text{alc. KOH}}$ Major product :

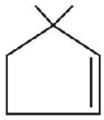
(a) 

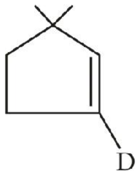
(b) 

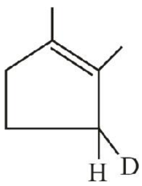
(c) 

(d) 

51.  $\xrightarrow[\Delta]{Ag_2O/H_2O}$ Major product :

(a) 

(b) 

(c) 

(d) $H_2C=CH_2$

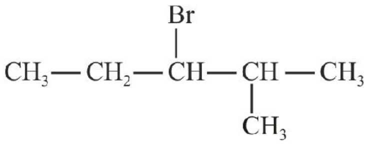
52. An halide $C_5H_{11}Br$ on treatment with alc. KOH give 2-pentene only. The halide will be:

(a) $CH_3-CH_2-CH_2-CH_2-CH_2-Br$

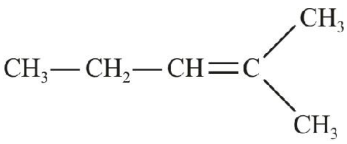
(b) $CH_3-CH_2-CH_2-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_3$

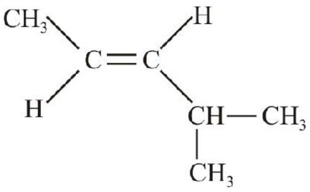
(c) $CH_3-CH_2-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2-CH_3$

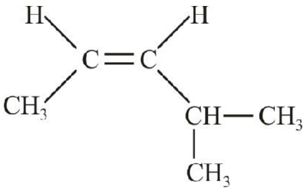
(d) $CH_3-CH-\overset{\displaystyle CH_3}{\overset{|}{\underset{\displaystyle Br}{\underset{|}{C}}}}-CH_3$

53. 

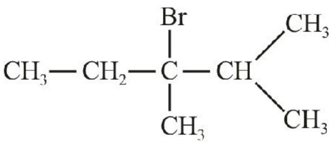
The major product obtained when this alkyl halide is subjected to E2 – reaction will be

(a) 

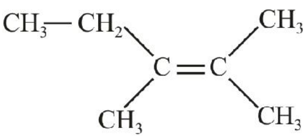
(b) 

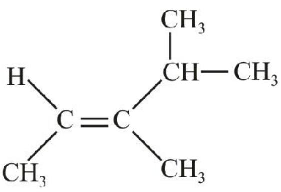
(c) 

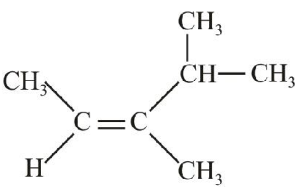
(d) All three products in equal proportions

54. 

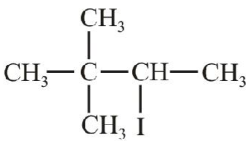
The major product obtained when this alkyl halide is subjected to E2 reaction will be

(a) 

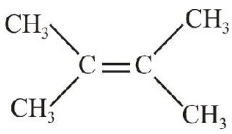
(b) 

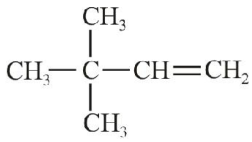
(c) 

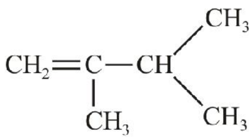
(d) all in equal proportion

55. 

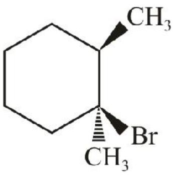
The major alkene obtained when this alkyl halide is subjected to E1 reaction will be

(a) 

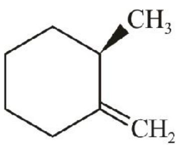
(b) 

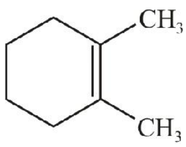
(c) 

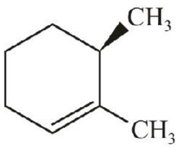
(d) All in equal proportions

56. 

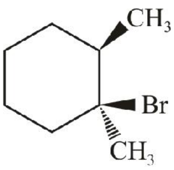
The major product obtained when this substrate is subjected to E1 reaction will

(a) 

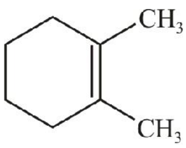
(b) 

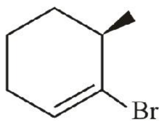
(c) 

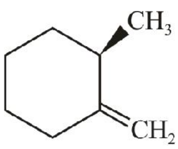
(d) none of these

57. 

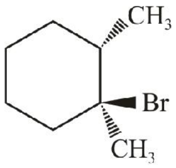
The major product obtained when this substrate is subjected to E2 reaction will be

(a) 

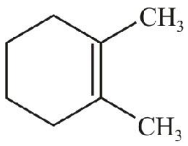
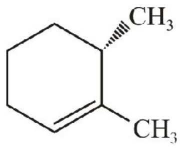
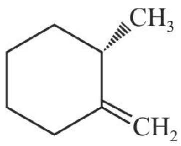
(b) 

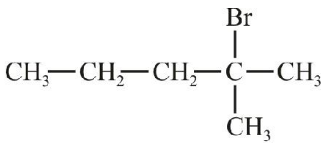
(c) 

(d) none of these

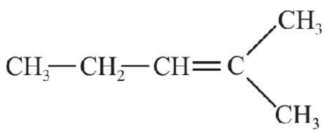
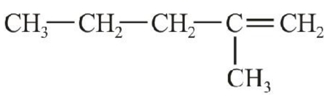
58. 

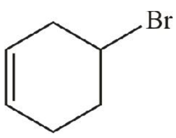
The major product obtained when this substrate is subjected to E2 reaction will be

(a) 
(b) 
(c) 
(d) none of these

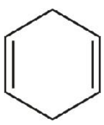
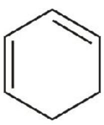
59. 

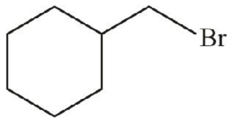
The major product obtained when this alkyl halide is subjected to E2 reaction under the treatment of potassium tert-butoxide will be

(a) 
(b) 
(c) both (a) and (b)
(d) none of these

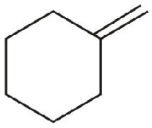
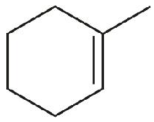
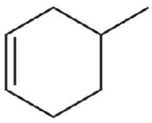
60. 

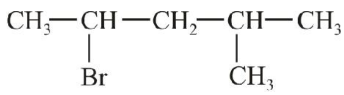
Which of the following products will be major if this substrate is subjected to E2 reaction?

(a) 
(b) 
(c) both in equal proportion
(d) not predictable

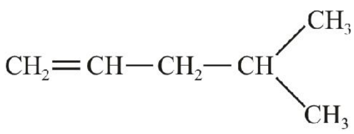
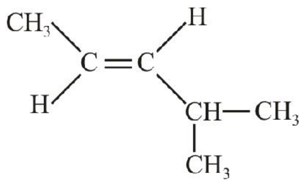
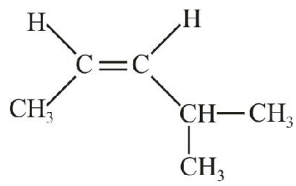
61. 

The products obtained when this substrate is subjected to E2 reaction will be

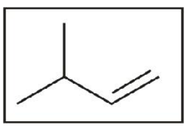
(a) 
(b) 
(c) 
(d) all of these

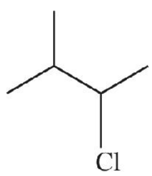
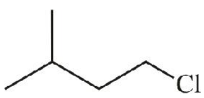
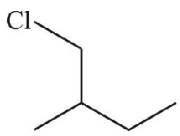
62. 

The major product obtained when this alkyl halide is subjected to E2 reaction will be

(a) 
(b) 
(c) 
(d) all three in equal proportions

63. What alkyl halide would give the following alkene as the only product in an elimination reaction ?



(a) 
(b) 
(c) 
(d) Both (a) and (b)

Substitution vs Elimination

64. S_N1 competes with E1 and S_N2 competes with E2. This is because

(a) both S_N1 and E1 have same rate-determining-step. Therefore E1 competes with S_N1.

(b) a base is a nucleophile and a nucleophile is a base, Therefore, S_N2 competes with E2.

(c) both a and b

(d) none of these

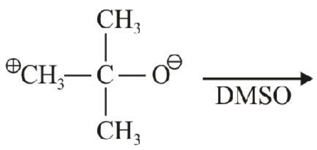
65. $CH_3-CH_2-CH_2-CH_2-Br + CH_3-CH_2-\bar{O} \xrightarrow{DMF}$

This reaction would follow which of the following pathway predominantly?

(a) S_N1 (b) S_N2

(c) E1 (d) E2

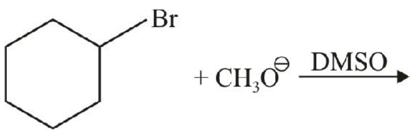
66. $CH_3-CH_2-CH_2-CH_2-Br$



This reaction would follow which of the following pathway predominantly?

(a) S_N1 (b) S_N2

(c) E1 (d) E2

67. 

This reaction would follow which of the following pathway predominantly?

(a) S_N1 (b) S_N2

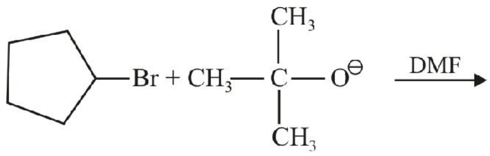
(c) S_N1/E1 (d) S_N2/E2

68. $CH_3-CH_2-CH_2-Br + \overset{\ominus}{I} \xrightarrow{DMSO}$

This reaction would follow which of the following pathway predominantly?

(a) S_N1 (b) S_N2

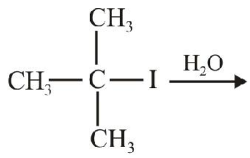
(c) E1 (d) E2

69. 

This reaction would follow which of the following pathway predominantly?

(a) S_N1 (b) S_N2

(c) E1 (d) E2

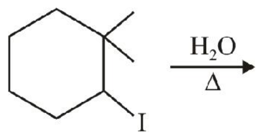
70. 

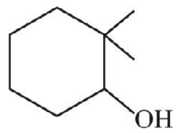
This reaction would follow which of the following pathways predominantly?

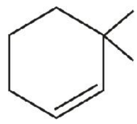
(a) S_N1/E1 (b) S_N2/E2

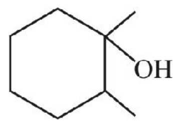
(c) S_N1/E2 (d) S_N2/E1

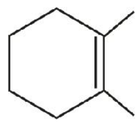
71. Find the major product of the following reaction :



(a) 

(b) 

(c) 

(d) 

72. If *tert*-butyl bromide and $NaNH_2$ reacts, the product formed is

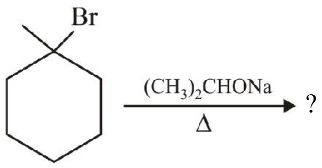
(a) *tert*-butylamine

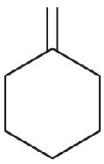
(b) *tert*-butylammonium bromide

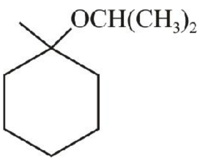
(c) a mixture of butyl amines

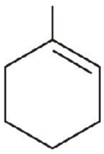
(d) *iso* butylene

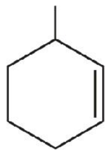
73. Choose the correct major product of the reaction:



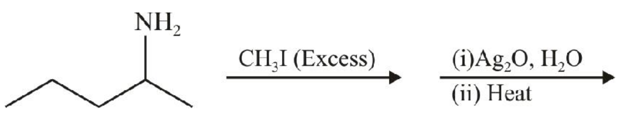
(a) 

(b) 

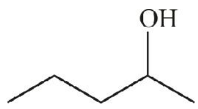
(c) 

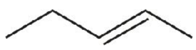
(d) 

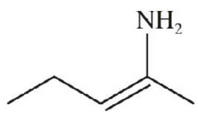
74. What is the major product of the reaction sequence?



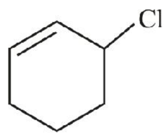
(a) 

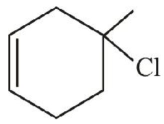
(b) 

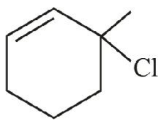
(c) 

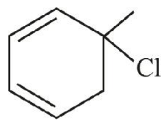
(d) 

75. Which of the following is the incorrect statement regarding the following four alkyl halides?

(I) 

(II) 

(III) 

(IV) 

(a) I is most reactive for a S_N2 reaction

(b) IV is most reactive for E1 reaction

(c) II is more reactive than III in S_N1 reaction

(d) IV is most reactive for E2 reaction

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTIONS

1. 2-chloro-2-methylpentane on with sodium methoxide in methanol yields: **(2016)**

(a) $C_2H_5CH_2\overset{\overset{\large CH_3}{|}}{\underset{\underset{\large CH_3}{|}}{C}}-OCH_3$

(b) $C_2H_5CH_2\underset{\underset{\large CH_3}{|}}{C}=CH_2$

(c) $C_2H_5CH=\underset{\underset{\large CH_3}{|}}{C}-CH_3$

(a) (a) and (c) (b) (c) only

(c) (a) and (b) (d) All of these

2. The organic chloro compound, which shows complete stereochemical inversion during an S_N2 reaction is **[2008]**

(a) $(C_2H_5)_2CHCl$ (b) $(CH_3)_3CCl$

(c) $(CH_3)_2CHCl$ (d) CH_3Cl

3. Elimination of bromine from 2-bromobutane results in the formation of **[2005]**

(a) predominantly 2-butyne

(b) predominantly 1-butene

(c) predominantly 2-butene

(d) equimolar mixture of 1 and 2-butene

4. $CH_3Br + Nu^- \rightarrow CH_3 - Nu + Br^-$

The decreasing order of the rate of the above reaction with nucleophile (Nu^-) A and D is:

[Nu = (A) PhO^-, (B) AcO^-, (C) HO^-, (D) CH_3O^-] **[2006]**

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

1. All questions marked **"S"** are single choice questions
2. All questions marked **"M"** are multiple choice questions
3. All questions marked **"C"** are comprehension based questions
4. All questions marked **"A"** are assertion–reason type questions

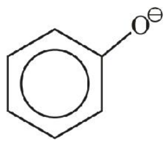
   (A) If both assertion and reason are correct and reason is the correct explanation of assertion.

   (B) If both assertion and reason are true but reason is not the correct explanation of assertion.

   (C) If assertion is true but reason is false.

   (D) If reason is true but assertion is false.
5. All questions marked **"X"** are matrix–match type questions
6. All questions marked **"I"** are integer type questions

Reagents

1. (S) (I)  (II) $CH_3-\overset{\overset{\displaystyle O}{||}}{C}-O^{\ominus}$

(III) $\overset{\ominus}{O}H$

Which of the following orders is correct for the nucleophilicity of these anions?

(a) I > II > III (b) III > II > I

(c) III > I > II (d) II > I > III

2. (S) $\overset{\ominus}{N}H_2$, $\overset{\ominus}{O}H$, NH_3, $\overset{\ominus}{F}$

Among these nucleophiles, nucleophilicity order is?

(a) $\overset{\ominus}{N}H_2 > \overset{\ominus}{O}H > \overset{\ominus}{F} > NH_3$

(b) $\overset{\ominus}{N}H_2 > \overset{\ominus}{O}H > NH_3 > \overset{\ominus}{F}$

(c) $\overset{\ominus}{N}H_2 > NH_3 > \overset{\ominus}{O}H > \overset{\ominus}{F}$

(d) None of these

3. (S) $\overset{\ominus}{F}$, NH_3, H_2O

Among these nucleophiles, nucleophilicity order is

(a) $\overset{\ominus}{F} > NH_3 > H_2O$ (b) $NH_3 > H_2O > \overset{\ominus}{F}$

(c) $NH_3 > \overset{\ominus}{F} > H_2O$ (d) none of these

4. (S) Which of the following species is an electrophile?

(a) BH_3 (b) $\overset{\oplus}{N}O_2$

(c) $\overset{\oplus}{Br}$ (d) all of these

5. (M) Which of the following are electrophiles?

(a) BF_3 (b) $:CCl_2$

(c) $\overset{\oplus}{N}H_4$ (d) $I^{\ominus}$

6. (M) In which of following pairs the first one is the stronger base than second?

(a) $CH_3COO^{\ominus}$, $HCOO^{\ominus}$

(b) $\overset{\ominus}{O}H$, $\overset{\ominus}{N}H_2$

(c) $CH_2=\overset{\ominus}{C}H$ $HC\equiv\overset{\ominus}{C}$

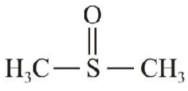
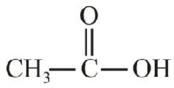
(d) CH_3NH_2, CH_3OH

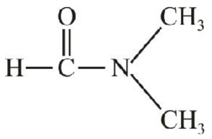
7. (M) Which of the following behave both as a nucleophile and an electrophile?

(a) $CH_3\ddot{N}H_2$ (b) CH_3-Cl

(c) $CH_3-C\equiv N$ (d) $CH_3-\overset{\overset{\displaystyle O}{||}}{C}-H$

8. (M) Which of the following solvents are aprotic?

(a)  (b) 

(c)  (d) H—S—H

9. (A) **Assertion (A):** $RS^{\ominus}$ is a stronger nucleophile and a better leaving group than $RO^{\ominus}$.

Reason (R): $RS^{\ominus}$ is a weaker base than $RO^{\ominus}$.

(a) A (b) B

(c) C (d) D

10. (A) **Assertion** : Nucleophilicity order in polar-protic solvent $I^{-} < Br^{-} < Cl^{-} < F^{-}$

Reason : Due to bigger size of $I^{\ominus}$ it is less solvated in polar-protic solvent.

(a) A (b) B

(c) C (d) D

11. (A) **Assertion :** Although flouride ion is a stronger base than iodide ion, iodide (I^{-}) is a better nucleophile than fluoride ion in aqueous solution.

Reason : Fluoride ion is more heavily hydrated than iodide ion.

(a) A (b) B

(c) C (d) D

12. (S) Arrange the following nucleophiles in the increasing order of nucleophilicity

(I) $(CH_3)_3CO^{-}$ (II) CH_3O^{-}

(III) cyclopentoxy anion

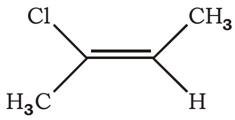
(IV) isopropoxy anion

(a) II < IV < III < I (b) IV < III < I < II

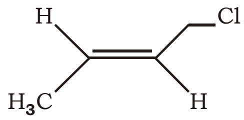
(c) III < IV < I < II (d) I < III < IV < II

Substitution

13. (M)



(I)



(II)

Choose the correct statements about molecule I and II.

(a) I and II are diastereomers.

(b) I and II are constitutional isomers.

(c) Reactivity for S_N1 path : II > I.

(d) Reactivity to S_N2 path : II > I.

14. (S) Which is the best solvent to use for the solvolysis reaction of t-butyl chloride ?

(a) Water (b) Carbon tetrachloride

(c) Formic acid (d) Heptane

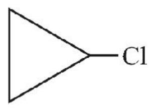
15. (S) Which of the following bases would give the best yield for the substitution product when reacted with 2-chloropropane ?

(a) CH_3COO^{-} (b) OH^{-}

(c) NH_2^{-} (d) $CH_3CH_2O^{-}$

16. (S) The compound most reactive towards S_N1 reaction is

(a) $Me_3C—CH_2Cl$ (b) $MeOCH_2Cl$

(c) $PhCH_2CH_2Cl$ (d) 

17. (M) Which of the following phrases are correctly linked with S_N1 reaction ?

(a) Rate is affected by polarity of solvent

(b) Rearrangement is possible

(c) The strength of the nucleophile is important in determining rate

(d) The reactivity series is tertiary > secondary > primary

18. (A) **Assertion :** Benzyl bromide when kept in acetone - water, produces benzyl alcohol.

Reason : The reaction follows S_N2 mechanism.

(a) A (b) B

(c) C (d) D

19. (A) **Assertion :** Rate of reaction is dependent only on the concentration of nucleophile in S_N1 reactions.

Reason : Polar solvent favours S_N1 reaction.

(a) A (b) B

(c) C (d) D

20. (A) **Assertion :** S_N2 reaction of CH_3—Br is faster in DMSO than in H_2O.

Reason : DMSO has greater capability to solvate nucleophile.

(a) A (b) B
(c) C (d) D

21. (A) **Assertion :** In S_N1 mechanism, the product with inversion of configuration is obtained in higher amount compared to the product with the retention of configuration.

Reason : Front side attack of nucleophile is hindered due to the presence of leaving group in the vicinity.

(a) A (b) B
(c) C (d) D

22. (S) Which of the following statements is correct about retention/invertion of configuration during the course of a reaction?

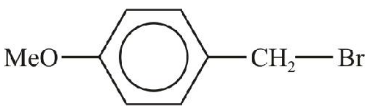
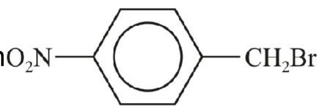
(a) Configuration is retained if reagent attacks from the same side of leaving group.

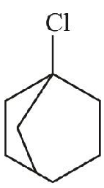
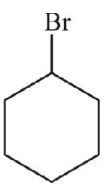
(b) Configuration is inverted if reagent attacks from the backside of leaving group.

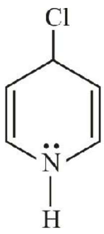
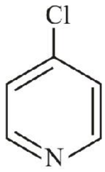
(c) No change in configuration is possible if no bonds of the asymmetric centre are cleaved during the course of reaction.

(d) all of these

23. (M) Consider the following statements and pick up the correct statements:

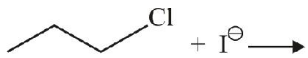
(a)  will react more readily than  for S_N1 reaction

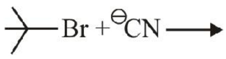
(b)  will react more readily than  for S_N1 reaction

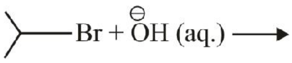
(c)  will react more readily than  for S_N1 reaction

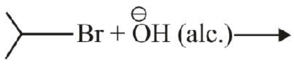
(d) S_N1 reaction occurs in polar protic solvent

24. (M) Which of the following are S_N2 reactions ?

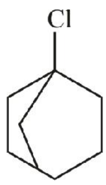
(a) 

(b) 

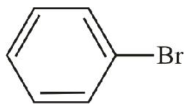
(c) 

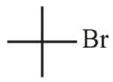
(d) 

25. (M) Which of the following compounds will not give S_N2 reaction?

(a) 

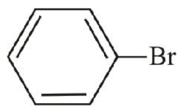
(b) $H_2C{=}CH{-}Cl$

(c) 

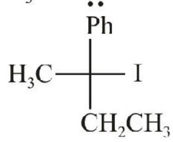
(d) 

26. (M) Which of the following compounds will give S_N1 reaction?

(a) $H_3C{-}CH_2{-}Br$

(b) 

(c) $H_3C{-}\ddot{O}{-}CH_2{-}Cl$

(d) 

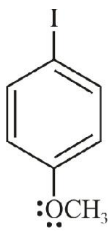
Comprehenstion

Aliphatic nucleophilic substitution is mainly of two type S_N1 and S_N2. S_N2 reaction proceed with strong nucleophile in polar aprotic solvent 3° halides do not give S_N2 reaction. Inverted products are obtained in this reaction and mechanism of reaction occurs through the formation of transition state.

S_N1 reaction proceed through the formation of carbocation in polar aprotic solvent. Solvent itself acts as nucleophile in this reaction. Racemization takes place in S_N1 reaction.

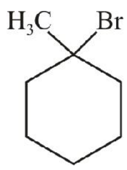
27. (C) Which of the following compounds will S_N1 reaction?

(a) $Ph-C(CH_3)(Cl)-CH_2CH_3$ (b) H_3C-Cl

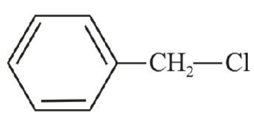
(c) H_3C-CH_2-Br (d) 

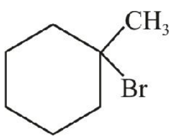
28. (C) Which one of the following will give racemised product in C_2H_5OH?

(a) $H_3C-CH(CH_3)-Br$ (b) $Ph-C(CH_3)_2-Cl$

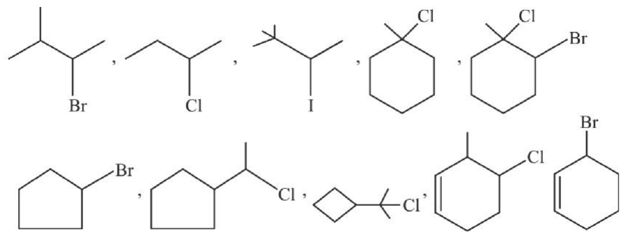
(c) $Ph-CH(CH_3)-Cl$ (d) 

29. (C) Which one of the following will give S_N2 reaction?

(a) $H_3C-C(CH_3)_2-Cl$ (b) 

(c)  (d) $H_2C=CH-Cl$

30. (I) How many substrates will show rearrangement during S_N1 reaction



31. (A) **Assertion :** 1° allylic halides are more reactive than 1° RX in S_N1 reaction.

Reason : Allylic carbocation intermediate is stabilised by resonance.

(a) A (b) B

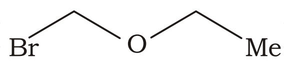
(c) C (d) D

32. (A) **Assertion :** Heavy metal ions Ag^+ or Pb^{2+} decrease S_N1 reactivity.

Reason : They aid ionisation of RX.

(a) A (b) B

(c) C (d) D

33. (A) **Assertion :** Rate of ethanolysis of 1° halide () by S_N1 mechanism is fast.

Reason : Carbocation is stabilised by resonance.

(a) A (b) B

(c) C (d) D

34. (A) **Assertion :** Treatment of (+)-2-chlorobutane with aqueous NaCN gives laevo rotatory product.

Reason : S_N2 reaction leads to inversion of configuration at the α-carbon.

(a) A (b) B

(c) C (d) D

35. (S) Consider the following nucleophiles

(I) $CF_3-SO_2-O^{\ominus}$ (II) $Ph-SO_2-O^{\ominus}$

(III) $Ph-O^{\ominus}$ (IV) $CH_3-CO-O^{\ominus}$

when attached to sp^3 -hybridized carbon, their leaving group ability in nucleophilic substitution reactions decreases in the order:

(a) I > II > III > IV (b) I > II > IV > III

(c) IV> I > II > III (d) IV > III > II > I

36. (S) The decreasing order of nucleophilicities of the following is:

(I) H_2O (II) EtOH

(III) $MeCOO^{\ominus}$ (IV) $\overset{\ominus}{O}H$

(V) $EtO^{\ominus}$

(a) V > IV > III > II >I (b) I > II > III > IV >V

(c) IV > V > III > II >I (d) I > II > III > V > IV

37. (S) Consider the following groups

(I) — $OCOCH_3$ (II) — OCH_3

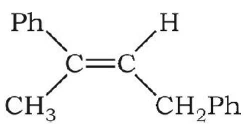
(III) — OSO_2CH_3 (IV) — OSO_2CF_3

The decreasing order of their leaving ability is

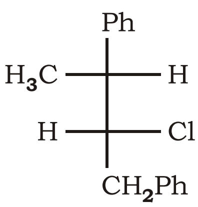
(a) I > II > III > IV (b) IV > III > I > II

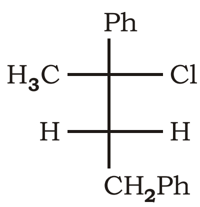
(c) III> II > I > IV (d) II > III > IV > I

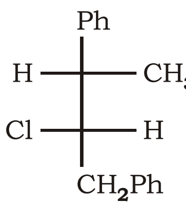
Elimination

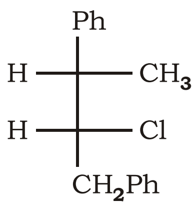
38. (M) Reactant $\xrightarrow[-HCl]{\text{alc. KOH (E2)}}$ 

Which of the following molecule(s) is/are suitable reactant ?

(a) 

(b) 

(c) 

(d) 

39. (S) $C_6H_6Cl_6$, on treatment with alcoholic KOH, yields

(a) C_6H_6 (b) $C_6H_3Cl_3$

(c) (C_6H_6) OH (d) $C_6H_6Cl_4$

40. (S) Debromination of mesodibromobutane will give the product as :

(a) n-butane (b) 2-butyne

(c) cis-2-butene (d) Trans-2-butene

41. (S) Among the following the most reactive towards alcoholic KOH is

(a) $CH_2 = CHBr$ (b) $CH_3COCH_2CH_2Br$

(c) CH_3CH_2Br (d) $CH_3CH_2CH_2Br$

42. (S) $CH_3CD_2CHBrCH_2CD_3$ on reaction with alc. KOH gives

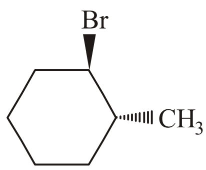
(a) $CH_3CD_2CH{=}CHCD_3$

(b) $CH_3CD{=}C{=}CHCD$

(c) $CD_3CD_2CH{=}CHCH_3$

(d) $CH_3CD{=}CHCH_2CD_3$

43. (S) What is the product of the reaction of the following compound with alcoholic potassium hydroxide ?



(a) 1-methylcyclohexane only

(b) 1-Methylcyclohexene only

(c) 3-methylcyclohexene (major product),

1-methylcyclohexene (minor product)

(d) 3-methylcyclohexene only

44. (S) Which of the following undergoes E2 elimination in the presence of a strong base to yield one product ?

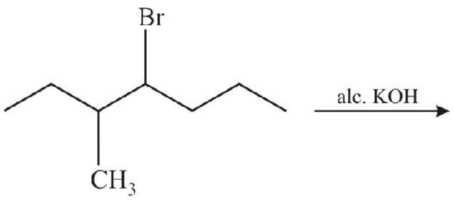
(a) 3-bromo-2-methylpentane

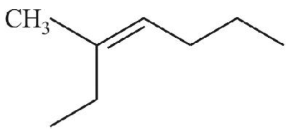
(b) 1-bromo-1-methylcyclohexane

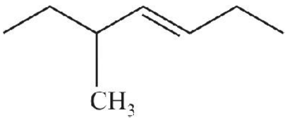
(c) 1-bromo-3, 3-dimethylbutane

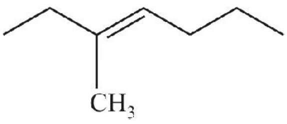
(d) 3-bromo-3-methylpentane

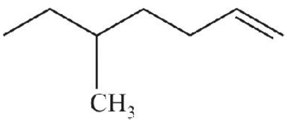
45. (M) In the given reaction the products formed can be



(a) 

(b) 

(c) 

(d) 

46. (M) Which of the following statements correctly describes(s) E1 reactions of alkyl halides (RX) ?

(a) Rearrangements are sometimes seen

(b) Rate = k [base] [RX]

(c) Rate = k [RX]

(d) The reactions occur in two or more distinct steps

47. (A) **Assertion :** 2-Bromobutane on reaction with sodium exthoxide in ethanol gives 1-butene as a major product.

Reason : 1-Butene is more stable than 2-butene.

(a) A (b) B

(c) C

(d) D (e) E

48. (A) **Assertion :** Tertiary haloalkanes are more reactive than primary haloalkanes towards elimination reactions.

Reason : The + I-effect of the alkyl groups weakens the C–X bond.

(a) A (b) B

(c) C (d) D

49. (X)

Column I	Column II
(A) Anti-elimination	(p) E1
(B) Must have acidic β-hydrogen	(q) E_i
(C) No effect of solvent	(r) E2
(D) carbocation	(s) E1 CB

Comprehension

An alkyl halide with β-hydrogen atoms when reacted with a base or a nucleophile has two competing routes : Substitution (S_N1 and S_N2) and elimination. Which route will be taken up depends upon the nature of the alkyl halide, strength and size of the base/nucleophile and reaction conditions. Thus, bulkier nucleophile and reaction conditions. Thus, bulkier nucleophile prefers to act as a base and abstracts a proton rather than approaching a tetravalent carbon atom (steric reasons) and vice-versa.

50. (C) Isopropyl bromide on heating with a concentrated solution of alcoholic (ethanolic) KOH predominantly gives

(a) Propene (b) Propan-2-ol

(c) Propan-1-ol (d) Isopropyl ethyl ether

51. (C) 2-Bromopropane is separately heated with aq. CH_3CO_2Na or with CH_3CH_2ONa/CH_3CH_2OH, the major product obtained in each case respectively are

(a) Propene, isopropyl ethyl ether

(b) Isopropyl acetate, propene

(c) Isopropyl acetate, isopropyl ethyl ether

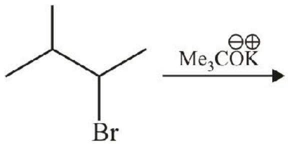
(d) Propene in both the cases

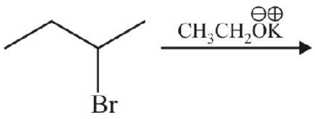
52. (C) 2-Bromopentane is heated with potassium ethoxide in ethanol. The major product obtained is

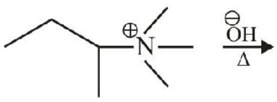
(a) 2-Ethoxypentane (b) Pentene-1

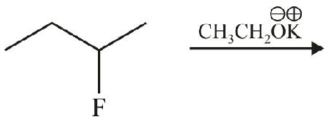
(c) cis-Pentene-2 (d) trans-Pentene-2

53. (M) In which product formation takes place according to Hofmann's rule?

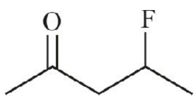
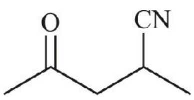
(a) 

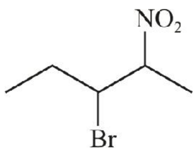
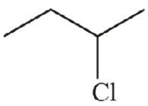
(b) 

(c) 

(d) 

54. (M) Which of the following can give E_1 cb reaction?

(a)  (b) 

(c)  (d) 

Comprehension

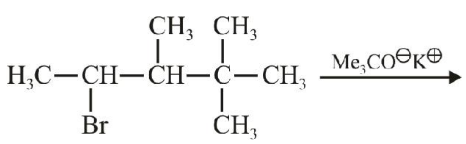
Type of elimination reaction in which least substituted alkene is major product known as Hofmann's elimiation. Such reaction occur in following conditions :

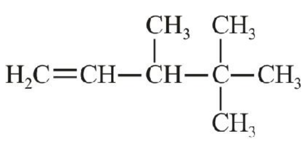
(X) when base is bulky

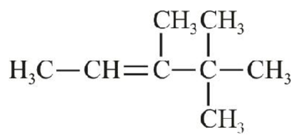
(Y) when leaving group is very poor such as fluoride, ammonium group ($-\overset{\oplus}{N}R_3$) etc.

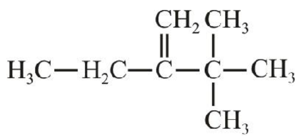
(Z) when alkyl halide contain one or more double bonds.

55. (C) What is the major product of the following reaction?



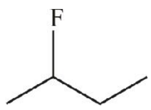
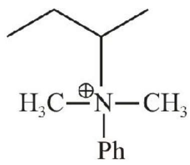
(a) 

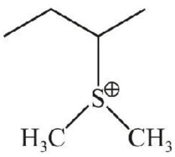
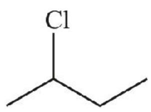
(b) 

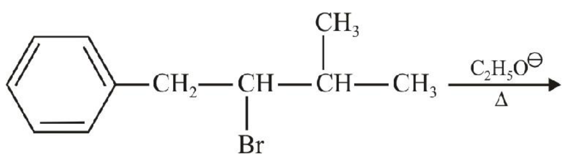
(c) 

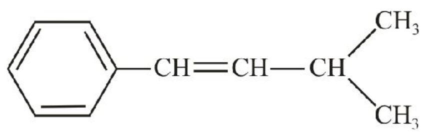
(d) None of these

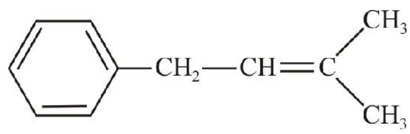
56. (C) Which of the following will not produce Hofmann's alkene as major product on reaction with strong base?

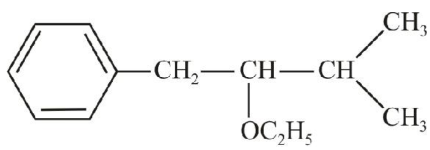
(a)  (b) 

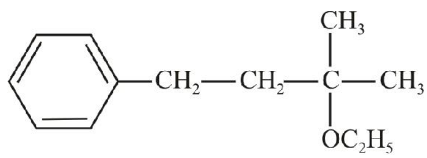
(c)  (d) 

57. (C) 

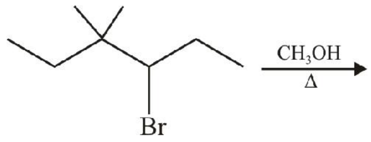
(a) 

(b) 

(c) 

(d) 

58. (I) Find out numbers of possible E_1 products from following reaction.



59. (A) **Assertion :** E_1cb reaction is favoured by stabilisation of carbanion and poor leaving group.

Reason : The reaction is kinetically of the second order and unimolecular.

(a) A (b) B

(c) C (d) D

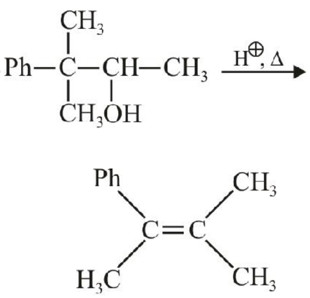
60. (A) **Assertion :** Ph–CO–$CH_2CH_2OCH_3$ has greater reactivity for E1CB than for E2 reaction.

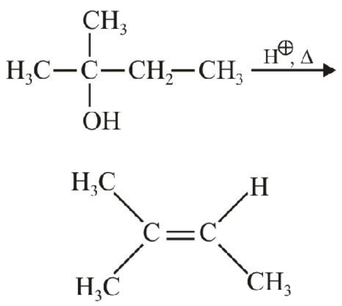
Reason : A poor leaving group and acidic β—H favour E1CB mechanism.

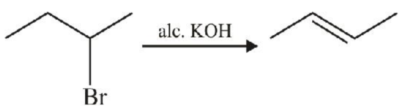
(a) A (b) B

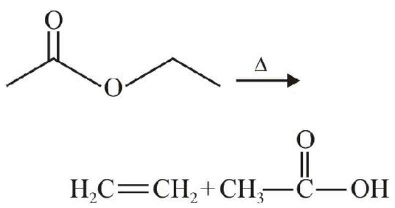
(c) C (d) D

61. (X) **Column (I)** **Column (II)**

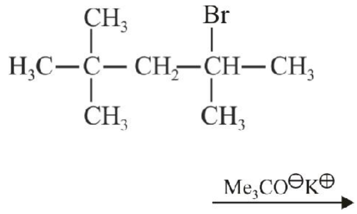
(A)  (p) Elimination

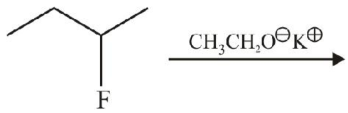
(B)  (q) Rearrangement

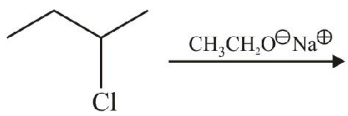
(C)  (r) Unimolecular

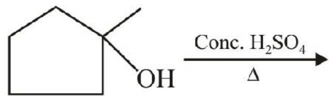
(D)  (s) Bimolecular

62. (X) **Column (I)** **Column (II)**

(A)  (p) Saytzeff's product

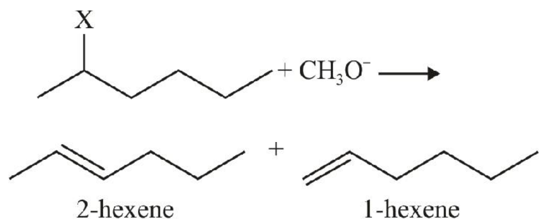
(B)  (q) Carbocation

(C)  (r) E_2 elimination

(D)  (s) Hofmann product

Comprehension

The following E2 is carried out with different halogen substituent



Four different reaction flasks, containing equimolar concentration of sodium methoxide in methanol solvent, were charged with equal moles of one of the four halide in each flask and allowed to react under identical experimental conditions as shown above. After a week, the percentage yield of two alkenes in each flask were measured and listed below

Flask No.	Leaving group	Conj. Acid pKa	% Yield of 2-hexene	% Yield of 1-hexene
1.	I^-	–10	81	19
2.	Br^-	–9	72	28
3.	Cl^-	–7	67	33
4.	F^-	3.2	30	70

From the percentage yield of two products observed with different leaving group, it was concluded that both, ease of leaving group and steric hindrance to the approach of base CH_3O^- to β–H control the orientation of elimination reaction.

63. (C) Based on the above observation, which flask will have the largest amount of 1-hexene?

(a) 1 (b) 2

(c) 3 (d) 4

64. (C) Transition state with maximum double-bond character will be formed when methoxide reacts with

(a) 2-iodohexane to form 2-hexene

(b) 2-iodohexane to form 1-hexene

(c) 2-fluorohexane to form 2-hexene

(d) 2-fluorohexane to form 1-hexene

65. (C) Transition state with least double-bond character will be formed when methoxide reacts with

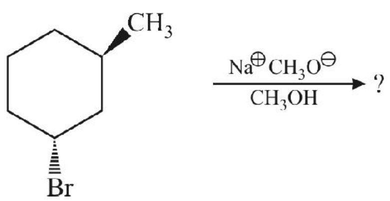
(a) 2-iodohexane to form 2-hexene

(b) 2-iodohexane to form 1-hexene

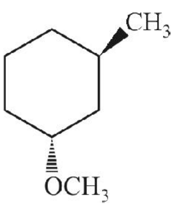
(c) 2-fluorohexane to form 2-hexene

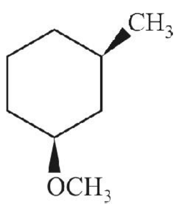
(d) 2-fluorohexane to form 1-hexene

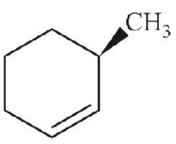
Substitution vs Elimination

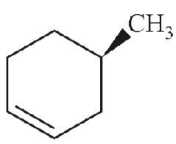
66. (M) 

Here the product can be ?

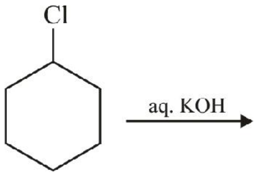
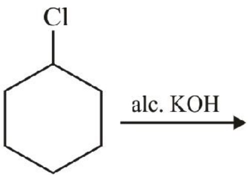
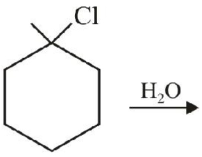
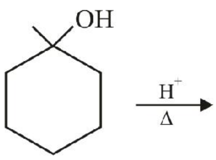
(a) 

(b) 

(c) 

(d) 

67. (X) Match the column (I) and (II)

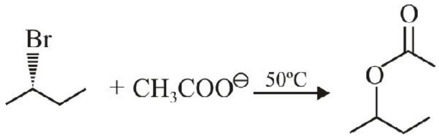
Column (I) Reaction	Column (II) Type of Reaction
(A) 	(p) S_N1
(B) 	(q) S_N2
(C) 	(r) E_1
(D) 	(s) E_2

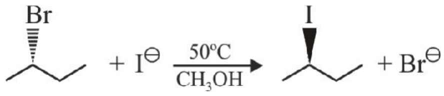
68. (I) If 1-chloro-4-methylcyclohexane is treated with KOH(*aq.*), both substitution (S_N2) and elimination (E2) reactions occur simultaneously in comparable amount. In total, how many substitution plus elimination products would result ?

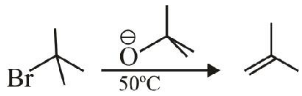
69. (I) 1, 3-dichlorocyclopentane exists in three stereo-isomeric forms, of which only two are chiral. If one mole of its achiral stereoisomer is treated with exactly one mole of NaOH in aqueous medium and, only mono substitution (S_N2) and mono-elimination (E2) occurs producing chlorocyclopentanol and chlorocyclopentane, how many different products are expected?

70. (S) Choose the answer that incorrectly shows the major product of the reaction.

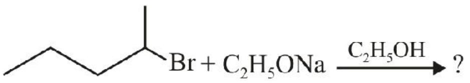
(a) 

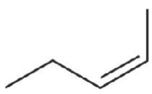
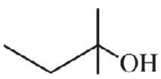
(b) 

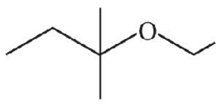
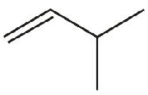
(c) 

(d) 

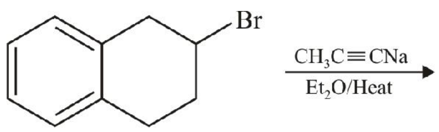
71. (S) What is the major product of the following reaction?

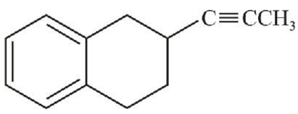
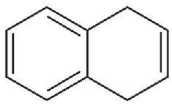


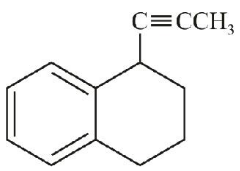
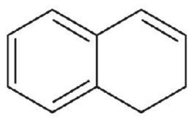
(a)  (b) 

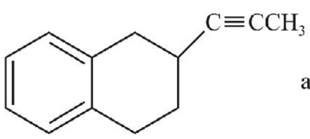
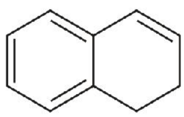
(c)  (d) 

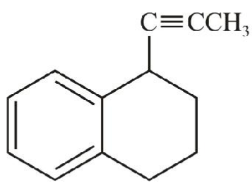
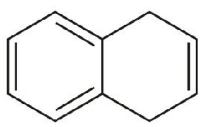
72. (S) In the following reaction, the predominant substitution and elimination products are



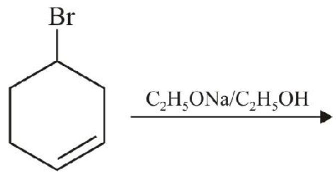
(a)  and 

(b)  and 

(c)  and 

(d)  and 

73. (S) What is true about the following reaction?



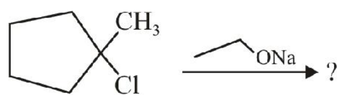
(a) Only substitution product is formed.

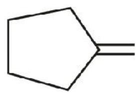
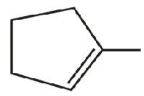
(b) Only elimination product is formed.

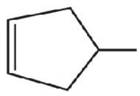
(c) Both substitution and elimination products are formed and substitution dominates.

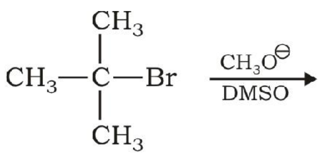
(d) Both substitution and elimination products are formed and elimination dominates.

74. (S) Choose the major product of the reaction.



(a)  (b) 

(c)  (d) 

75. (S) 

This reaction would follow which of the following pathway predominantly ?

(a) S_N1 (b) S_N2

(c) E1 (d) E2

EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

Objective Questions

1. Which of the following has the highest nucleophilicity ? **[2000]**

(a) F^- (b) OH^-

(c) CH_3^- (d) NH_2^-

2. The order of reactivities of the following alkyl halides for a S_N2 reaction is **[2000]**

(a) RF > RCl > RBr > RI

(b) RF > RBr > RCl > RI

(c) RCl > RBr > RF > RI

(d) RI > RBr > RCl > RF

3. An S_N2 reaction at an asymmetric carbon of a compound always gives **[2001]**

(a) an enantiomer of the substrate

(b) a product with opposite optical rotation

(c) a mixture of diastereomers

(d) a single steroisomer

True/False

4. Iodide is better nucleophile than bromide. **[1985]**

5. During S_N1 reactions, the leaving group leaves the molecule before the incoming group is attached to the molecule. **[1990]**

Subjective Questions

6. Arrange the following in ordre of their

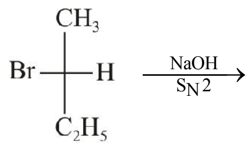
(i) Increasing basicity

H_2O, OH^-, CH_3OH, CH_3O^-

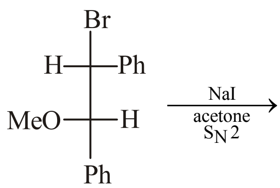
(ii) Increasing reactivity in nucleophilic substitution re actions

CH_3F, CH_3I, CH_3Br, CH_3Cl **[1992]**

7. Draw the stereochemcial structure of product in the following reaction. **[1994]**



8. Predict the structure of the product in the following reaction



[1996, 1M]

9. Predict the structures of the products:

$C_6H_5CH_2CHClC_6H_5 \xrightarrow{\text{Alc. KOH}}$ **[1998, 2M]**

ANSWER KEY

EXERCISE - 1 : (Basic Objective Questions)

1. (a)	**2.** (a)	**3.** (a)	**4.** (a)	**5.** (c)	**6.** (a)	**7.** (a)	**8.** (a)	**9.** (b)	**10.** (b)
11. (b)	**12.** (a)	**13.** (c)	**14.** (c)	**15.** (d)	**16.** (a)	**17.** (d)	**18.** (b)	**19.** (d)	**20.** (a)
21. (a)	**22.** (a)	**23.** (a)	**24.** (c)	**25.** (b)	**26.** (b)	**27.** (d)	**28.** (c)	**29.** (c)	**30.** (d)
31. (a)	**32.** (d)	**33.** (d)	**34.** (b)	**35.** (c)	**36.** (c)	**37.** (c)	**38.** (c)	**39.** (c)	**40.** (b)
41. (a)	**42.** (b)	**43.** (a)	**44.** (c)	**45.** (b)	**46.** (a)	**47.** (b)	**48.** (d)	**49.** (b)	**50.** (b)
51.(d)	**52.** (c)	**53.** (a)	**54.** (a)	**55.** (a)	**56.** (b)	**57.** (a)	**58.** (b)	**59.** (b)	**60.** (b)
61. (a)	**62.** (b)	**63.** (b)	**64.** (c)	**65.** (b)	**66.** (d)	**67.** (d)	**68.** (b)	**69.** (d)	**70.** (a)
71. (d)	**72.** (d)	**73.** (a)	**74.** (a)	**75.** (c)					

EXERCISE - 2 : (Previous Year JEE Mains Questions)

1. (d) **2.** (d) **3.** (c) **4.** D > C > A > B

EXERCISE - 3 : (Advanced Objective Questions)

1. (c)	**2.** (b)	**3.** (c)	**4.** (d)	**5.** (ab)	**6.** (acd)	**7.** (cd)	**8.** (ac)	**9.** (b)	**10.** (d)
11. (a)	**12.** (d)	**13.** (bcd)	**14.** (a)	**15.** (a)	**16.** (b)	**17.** (abd)	**18.** (c)	**19.** (d)	**20.** (c)
21. (a)	**22.** (d)	**23.** (acd)	**24.** (ac)	**25.** (abcd)	**26.** (cd)	**27.** (a)	**28.** (c)	**29.** (b)	**30.** (0006)
31. (a)	**32.** (d)	**33.** (a)	**34.** (d)	**35.** (b)	**36.** (a)	**37.** (b)	**38.** (abc)	**39.** (b)	**40.** (d)
41. (b)	**42.** (a)	**43.** (d)	**44.** (c)	**45.** (abc)	**46.** (acd)	**47.** (e)	**48.** (b)		

49. A–r,s; B–s, C–q, D–p **50.** (a) **51.** (b) **52.** (d) **53.** (acd) **54.** (ab) **55.** (a) **56.** (d)

57. (a) **58.** (0004) **59.** (b) **60.** (a) **61.** (A – p, q, r; B – p, r; C – p, s; D – p, r)

62. (A – r, s; B – r, s; C – p, r; D – p, q) **63.** (d) **64.** (a) **65.** (c) **66.** (bcd)

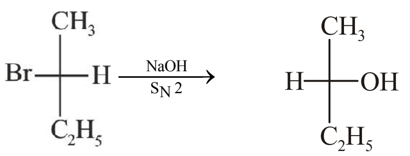
67. (A – q; B – s; C – p; D – r) **68.** (0004) **69.** (0004) **70.** (a) **71.** (a) **72.** (c) **73.** (d) **74.** (b)

75. (d)

EXERCISE - 4 : (Previous Year JEE Advanced Questions)

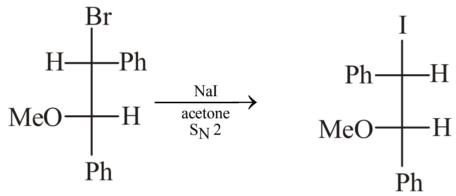
1. (c) **2.** (d) **3.** (d) **4.** True **5.** True **6.** $H_2O < CH_3OH < HO^- < CH_3O^-$

7.



product with inverted configuration

8.



inversion of configuration occur only at α-carbon

9.